MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, ADVANCED SCHOTTKY TTL,

EXCLUSIVE OR GATES, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, Advanced Schottky TTL, exclusive OR gates microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Quadruple, two-input exclusive OR gate

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Letter	Case outline (see MIL-M-38510, appendix C)
A	F-1 (14-lead, 1/4" x 1/4"), flat package
B	F-3 (14-lead, 3/16" x 1/4"), flat package
C	D-1 (14-lead, 1/4" x 3/4"), dual-in-line package
D	F-2 (14-lead, 1/4" x 3/8"), flat package
2	C-2 (20 terminal, .350" x .350"), square chip carrier package
X	C-2A (20 terminal, .350" x .350"), square chip carrier package

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to +7.0 V dc
Input voltage range	-1.2 V dc at -18 mA to +7.0 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation (P_D) 1/	154 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction to case (θ_{JC}):	
Cases A, B, C, D, 2, and X	(See MIL-M-38510, appendix C)
Junction temperature (T_J) 2/	+175°C

1/ Must withstand added P_D due to short circuit test (e.g., I_{OS}).
2/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions in accordance with method 5004 of MIL-STD-883.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Supply voltage - - - - - - - - - - - - - - -	4.5 V dc to 5.5 V dc
Minimum high level input voltage - - - -	2.0 V dc
Maximum low level input voltage- - - - -	0.8 V dc
Case operating temperature range (T_C)- -	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections and logic diagram. The terminal connections and logic diagram shall be as specified on figure 1.

3.2.2 Truth table and logic equation. The truth table and logic equation shall be as specified on figure 2.

3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.4).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$ unless otherwise specified	Device type	Limits		Unit
				Min	Max	
High level output voltage	V_{OH}	V_{CC} = 4.5 V; V_{IL} = 0.8 V; I_{OH} = -1 mA; V_{IH} = 2.0 V	01	2.5		V
Low level output voltage	V_{OL}	V_{CC} = 4.5 V; V_{IH} = 2.0 V; I_{OL} = 20 mA; V_{IL} = 0.8 V	01		0.5	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V; I_{IN} = -18 mA; T_C = +25°C	01		-1.2	V
High level input current	I_{IH1}	V_{CC} = 5.5 V; V_{IH} = 2.7 V	01		20	µA
	I_{IH2}	V_{CC} = 5.5 V; V_{IH} = 7.0 V	01		100	µA
Low level input current	I_{IL}	V_{CC} = 5.5 V; V_{IL} = 0.5 V	01	-.03	-.60	mA
Short-circuit output current 1/	I_{OS}	V_{CC} = 5.5 V; V_{OS} = 0 V	01	-60	-150	mA
Output drive	I_{OD}	V_{CC} = 4.5 V; V_{IN} = 5.5 V V_{OUT} = 2.5 V	01	60	-	mA
Supply current	I_{CC1}	V_{CC} = 5.5 V; V_{IN} = 0 V	01		23	mA
	I_{CC2}	V_{CC} = 5.5 V; V_{IN} = 5.5 V	01		28	mA
	I_{CCH}	V_{CC} = 5.5 V	01		18	mA
Propagation delay time, low-to-high level		V_{CC} = 5.0 V C_L = 50 pF R_L = 500Ω				
A, B, to Y	t_{PLH1}		01	1.0	8.5	ns
	t_{PLH2}		01	1.0	7.0	ns
Propagation delay time, high-to-low level		V_{CC} = 5.0 V C_L = 50 pF R_L = 500Ω				
A, B, to Y	t_{PHL1}		01	1.0	8.0	ns
	t_{PHL2}		01	1.0	7.0	ns

1/ Not more than one output should be shorted at a time.

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TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III) Class S devices	Class B devices
Interim electrical test parameters (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,7,9, 10,11	1*,2,3,7,9
Group A electrical test requirements (method 5005)	1,2,3,7,8, 9,10,11	1,2,3,7,8,9, 10,11
Group B electrical test requirements (method 5005) subgroup 5	1,2,3,9,10, 11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

*PDA applies to subgroup 1 (see 4.2c).

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 8 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspections. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 3, or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

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4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical parameters shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions, or equivalent:

(1) Test condition D or E, using the circuit shown on figure 3, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein:

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

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Device type 01

Cases 2 and X

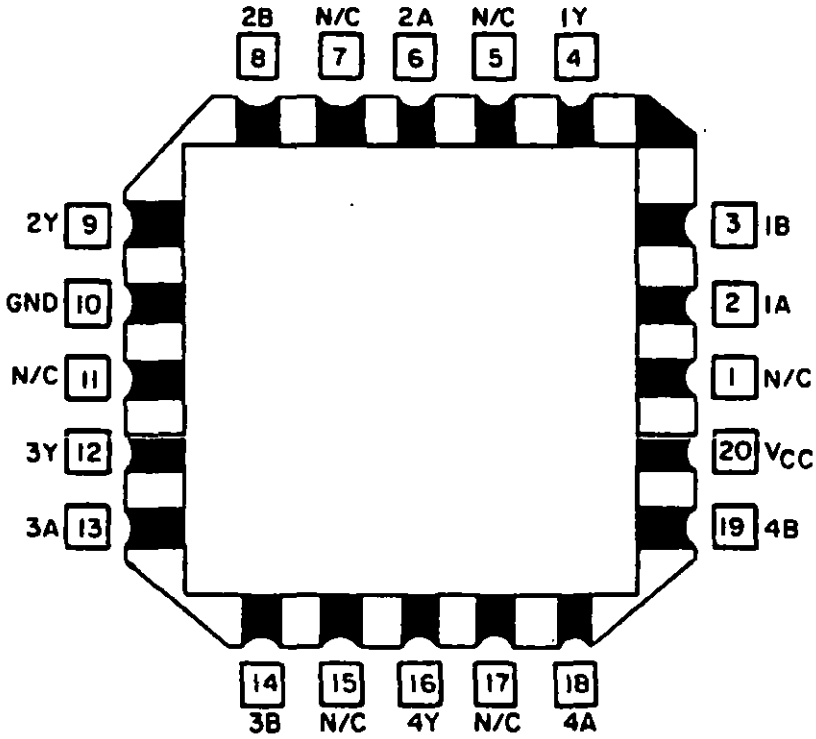


FIGURE 1. Terminal connections.

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Device Type 01

Cases A, B, C, and D

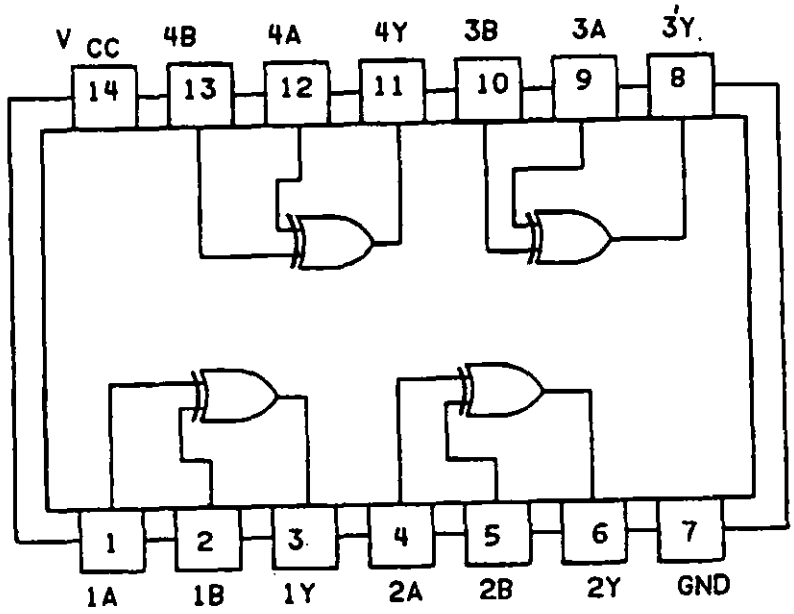


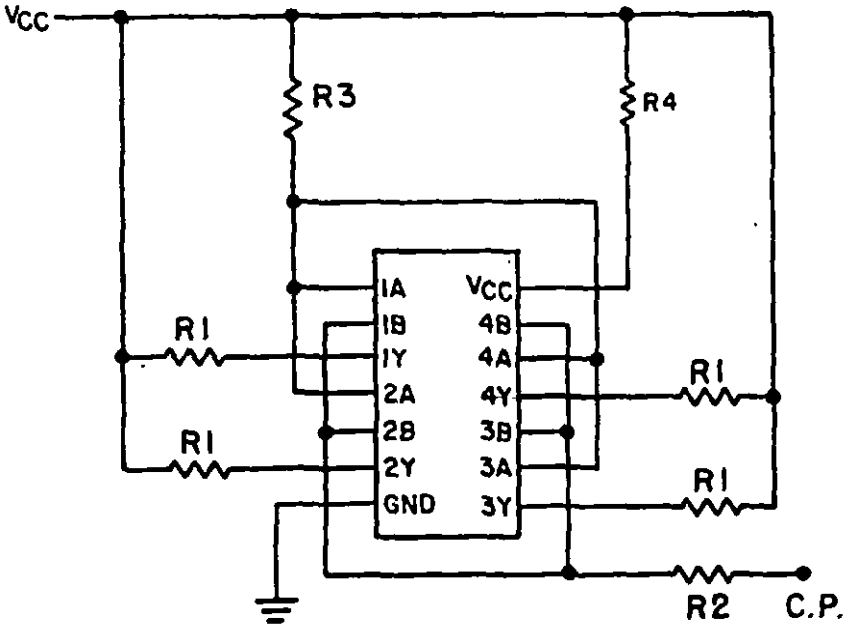
FIGURE 1. Terminal connections and logic diagram - Continued.

Inputs		Output
A	B	Y
L	L	L
L	H	H
H	L	H
H	H	L

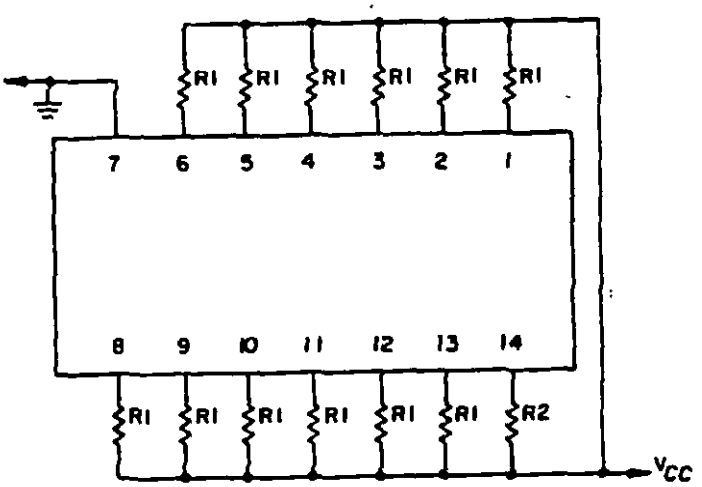
H = High level
L = Low level
Y = A ⊕ B

FIGURE 2. Truth table and logic equation.

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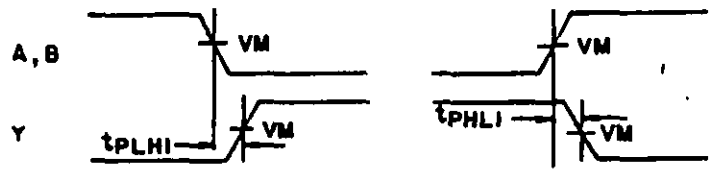
NOTES:
1. R1 = 240Ω ±5 percent; R2 = 51Ω ±5 percent.
   R3 = 1 kΩ ±5 percent.
2. V_{CC} and R4 shall be chosen to insure 5.5 V minimum is present at device V_{CC} terminal.
3. C.P or $\overline{C.P}$ = 100 kHz ±50 percent square wave; duty cycle = 50 ±15 percent; V_{IL} = -0.5 minimum to 0.8 V maximum; V_{IH} = 2.0 V to 5.5 V maximum.



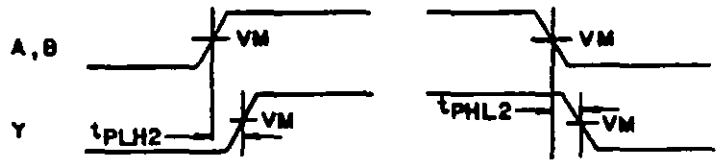
NOTES:
1. High temperature (175°C) condition F burn-in circuit only.
2. R1 = 2 kΩ, R2 = 2Ω.
3. V_{CC} shall be chosen such that 5.0 V minimum is present at device terminal.

FIGURE 3. Burn-in and life test circuit for device type 01.

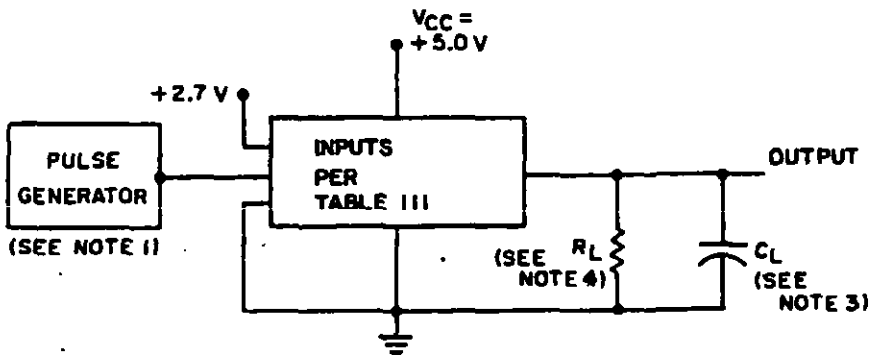
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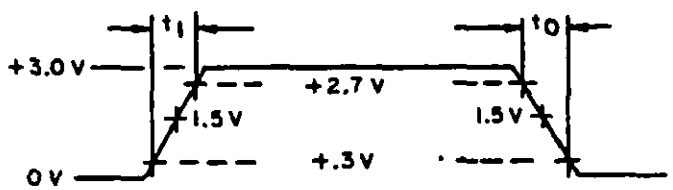


Device type 01



Device type 01





NOTES:
1. V_{IN} = Input pulse and has the following characteristics:

   PRR ≤ 1 MHz, $t_I = t_O \leq 2.5$ ns.
2. Inputs not under test are at ground or 2.7 V per table III.
3. C_L = 50 pF ±10 percent including scope probe, wiring and stray capacitance without package in test fixture.
4. R_L = 499 ±5 percent.
5. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 4. Switching time waveforms.

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TABLE III. Group A inspection for device type 01.

Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroups	Symbol	MIL-STD-883 method	Cases K, C, D 2, 3, 17/ Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	V_{OL}	3007	1 2 3	2.0 V	2.0 V	20 mA	2.0 V	2.0 V	20 mA	GND													4.5 V	01 02 03 04 05 06 07 08 09 10 11 12 13 14 15 16 17 18 19 20		0.5	V
	V_{OH}	3006	4 5 6 7 8 9 10 11 12	0.8 V 2.0 V	2.0 V 0.8 V	-1 mA -1 mA	0.8 V 2.0 V	2.0 V 0.8 V	-1 mA -1 mA																2.5		V
	V_{IC}		13 14 15 16 17 18 19 20	-18 mA	-18 mA		-18 mA	-18 mA															-1.2			V
	I_{IH1}	3010	21 22 23 24 25 26 27 28	2.7 V	2.7 V		2.7 V	2.7 V															5.5 V		1/		μA
	I_{IH2}		29 30 31 32 33 34 35 36	7.0 V	7.0 V		7.0 V	7.0 V															7.0 V		1/		μA
	I_{IL1}	3009	37 38 39 40 41 42 43 44	0.5 V	0.5 V		0.5 V	0.5 V															0.5 V		1/		μA
	I_{OS}	3011	45 46 47 48 49 50 51 52	5.5 V	0.0 V	0.0 V	5.5 V	0.0 V	0.0 V														5.5 V		-40	-150	mA
	I_{CC}	3005	53 54 55 56 57 58 59 60	5.5 V	0.0 V	2.5 V	5.5 V	0.0 V	2.5 V														5.5 V		60		mA
2, 3 T_C = +125°C and -55°C	V_{OL}	3007	1 2 3	2.0 V	2.0 V	20 mA	2.0 V	2.0 V	20 mA	GND													4.5 V	01 02 03 04 05 06 07 08 09 10 11 12 13 14 15 16 17 18 19 20		0.5	V
	V_{OH}	3006	4 5 6 7 8 9 10 11 12	0.8 V 2.0 V	2.0 V 0.8 V	-1 mA -1 mA	0.8 V 2.0 V	2.0 V 0.8 V	-1 mA -1 mA																2.5		V
	V_{IC}		13 14 15 16 17 18 19 20	-18 mA	-18 mA		-18 mA	-18 mA															-1.2			V
	I_{IH1}	3010	21 22 23 24 25 26 27 28	2.7 V	2.7 V		2.7 V	2.7 V															5.5 V		1/		μA
	I_{IH2}		29 30 31 32 33 34 35 36	7.0 V	7.0 V		7.0 V	7.0 V															7.0 V		1/		μA
	I_{IL1}	3009	37 38 39 40 41 42 43 44	0.5 V	0.5 V		0.5 V	0.5 V															0.5 V		1/		μA
	I_{OS}	3011	45 46 47 48 49 50 51 52	5.5 V	0.0 V	0.0 V	5.5 V	0.0 V	0.0 V														5.5 V		-40	-150	mA
	I_{CC}	3005	53 54 55 56 57 58 59 60	5.5 V	0.0 V	2.5 V	5.5 V	0.0 V	2.5 V														5.5 V		60		mA

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, C, D 2, 3 1/	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits Min	Limits Max	Unit
			Test No.	1A	1B	1C	2A	2B	2C	3D	3C	3A	3B	4C	4A	4B	V_{CC}				
1 T_C = +25°C	I_{CCL}	3005	54	5.5 V	5.5 V		5.5 V	5.5 V		Gnd		5.5 V	5.5 V		5.5 V	5.5 V	5.5 V	V_{CC}		28	mA
	I_{CCH}	3005	55	0.0 V	5.5 V		0.0 V	5.5 V		Gnd		0.0 V	5.5 V		0.0 V	5.5 V	5.5 V	V_{CC}		18	mA
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
7 T_C = +25°C	Functional tests 3/	3014	56 57 58 59	B A B A	B A A B	L L H H	B A B A	B A A B	L L H H	Gnd	L L H H	B A B A	B A A B	L L H H	B A B A	B A A B	4/	All outputs			
8	Repeat subgroup 7 test at T_C = +125°C and -55°C.																				
9 T_C = +25°C	t_{PLH1}	3003 (Fig. 4)	60 61 62 63 64 65 66 67	IN 2.7 V	2.7 V IN	OUT OUT	IN 2.7 V	2.7 V IN	OUT OUT	Gnd	OUT OUT	IN 2.7 V	2.7 V IN	OUT OUT	IN 2.7 V	2.7 V IN	0.0 V	1A to 1Y 1B to 2Y 2A to 2Y 2B to 3Y 3A to 3Y 3B to 4Y 4A to 4Y 4B to 4Y	1.5	7.0	ns
	t_{PHL1}		68 69 70 71 72 73 74 75	IN 2.7 V	2.7 V IN	OUT OUT	IN 2.7 V	2.7 V IN	OUT OUT		OUT OUT	IN 2.7 V	2.7 V IN	OUT OUT	IN 2.7 V	2.7 V IN				6.5	
	t_{PLH2}		76 77 78 79 80 81 82 83	IN 0.0 V	0.0 V IN	OUT OUT	IN 0.0 V	0.0 V IN	OUT OUT		OUT OUT	IN 0.0 V	0.0 V IN	OUT OUT	IN 0.0 V	0.0 V IN				5.5	
	t_{PHL2}		84 85 86 87 88 89 90 91	IN 0.0 V	0.0 V IN	OUT OUT	IN 0.0 V	0.0 V IN	OUT OUT		OUT OUT	IN 0.0 V	0.0 V IN	OUT OUT	IN 0.0 V	0.0 V IN					

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, C, D 2, 3, 17	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits Min	Limits Max	Units
			Cases E, X	1	3	4	6	8	9	10	12	13	14	16	18	19	20				
			Test no.	1A	1B	1C	2A	2B	2C	GND	3A	3B	3C	4A	4B	4C	V_{CC}				
10	t_{PLH1}	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C and limits as shown.																	1.0	8.5	ns
"	t_{PHL1}																		"	8.0	"
"	t_{PLH2}																		"	7.0	"
"	t_{PHL2}																		"	7.0	"
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																				

1/ For cases 2 and X, pins not referenced are N/C.

2/ I_{IL} limits shall be as follows:

Test	Min/Max limits in (mA) for circuit A	B	C
I_{IL1}	-.257/-.60	-.03/-.60	-.03/-.60

3/ A = 2.5 V; B = 0.5 V; H ≥2.6 V, L ≤0.5 V with high speed checker double comparator, or; H ≥1.5 V, L ≤1.0 V with high speed checker single comparator.

4/ Perform subgroup 7 sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

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6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND - - - - - - - - - - - - - Ground zero voltage potential.

I_{IN} - - - - - - - - - - - - - Current flowing into an input terminal.

V_{IN} - - - - - - - - - - - - - Voltage level at an input terminal.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2) lead material and finish C (see 3.3). Longer lead lengths and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54F86

6.6 Manufacturers' designation. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designation.

Device type	Circuits		
	A	B	C
	Fairchild	Motorola	Signetics
01	X		

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - SH, OS
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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